

Simon Garcia Salmeron

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EDUCATION

University of St. Thomas | St. Paul, MN | Expected May 2029

Bachelor of Science in Mechanical Engineering; Minor in Biomedical Engineering

GPA: 3.32/4.0 | Davis UWC Scholar

Relevant coursework: Mechanical Engineering Graphics (SolidWorks), Physiology

RESEARCH EXPERIENCE

Lee Research Group, University of St. Thomas | Jun. 2026-Present

Undergraduate Student Researcher | St. Paul, MN

Print optimization supporting rapid manufacturing of wearable biomedical strain sensors.

- Quantified pneumatic-dispensing extrusion (Nordson) across three nozzle diameters (265 measurements); at 3 wt% carbon nanotubes, a 0.51 mm nozzle, and 500 kPa, crosslinked ink flowed approximately three times faster than uncrosslinked ink after 30 minutes, returning toward baseline over six hours.
- Designed response surface experiments and fitted JMP models for line width ($R^2 = 0.81$) and width variability ($R^2 = 0.72$), evaluating ink loading, nozzle diameter, pressure, and print speed.
- Measured printed-line cross-sectional geometry using Keyence VHX-7000 microscopy and custom Python image analysis.
- Conducted a separate electrical-characterization experiment, measuring two-probe resistance in conductive samples with and without silver ink.

TECHNICAL PROJECTS

Lee Research Group Tool Suite | Jun. 2026-Present

- Developed a Python tool suite used by two researchers for Aerotech G-code workflows, cross-sectional image analysis, and printing-parameter prediction.
- Co-developed calibrated line-width analysis with a research partner, using automated edge detection and CSV measurement exports to support experimental analysis.

Aerodynamic Drag Analysis | Completed Apr. 2025

- Compared open- and closed-window vehicle drag at five airflow speeds using SimFlow/OpenFOAM, with Blender geometry edits and ParaView visualization.

ENGINEERING EXPERIENCE

Llansa Engineers | Mechanical Engineering Intern | Jan. 2026

Managua, Nicaragua

- Diagnosed mechanical and hydraulic faults and performed repairs on approximately five construction and transport machines.
- Restored a hydraulic hammer to construction-site operation and a small truck to material-transport service.

LEADERSHIP

UWC Mahindra College | Outdoor Educator & Team Leader | Sep. 2023-May 2025

- Coordinated safety and logistics for weekly outdoor excursions within a school community of 200+ members and instructed students in rappelling and cycling.

TECHNICAL SKILLS

Fabrication: Direct Ink Writing; Aerotech motion system; Nordson pneumatic dispensing

Characterization: Keyence VHX-7000 optical microscopy; cross-sectional analysis; two-probe resistance measurement

Experimental Design & Statistics: DOE, RSM, regression modeling, JMP

Programming & Analysis: Python, NumPy, OpenCV, G-code

CAD & Simulation: SolidWorks, SimFlow/OpenFOAM, Blender, ParaView